

Decision-making guide

E-commerce Microsimulation

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1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes market, Catalog, Marketing, Experience, Operations, Decision Checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

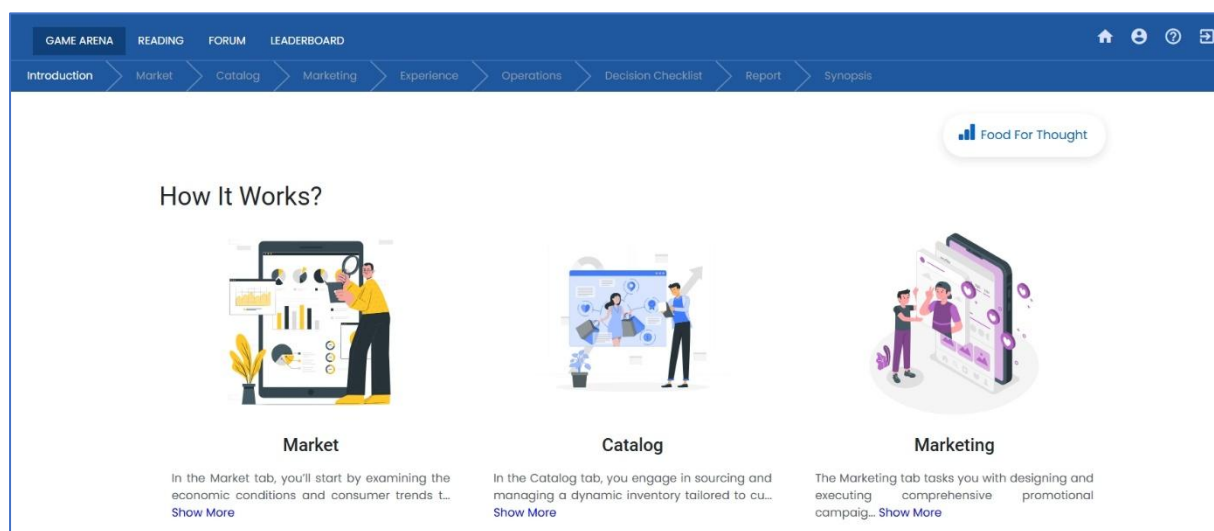


Figure 1 The Game Arena

1.2 Readings

This section provides essential materials necessary for the participants to actively participate in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the microsimulation. This video presents insights into the industry, ongoing trends, obstacles, and the diverse array of decisions required.

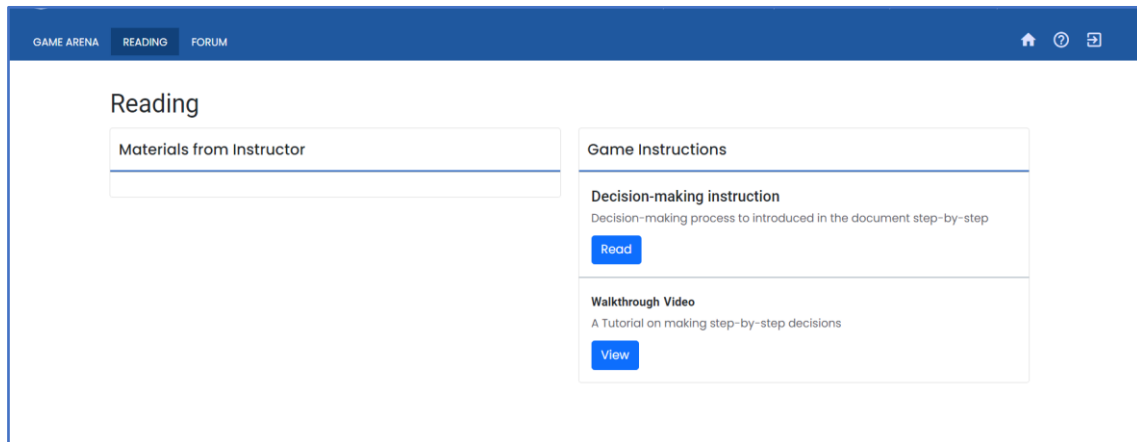


Figure 2 The Reading Tab

1.3 Forum

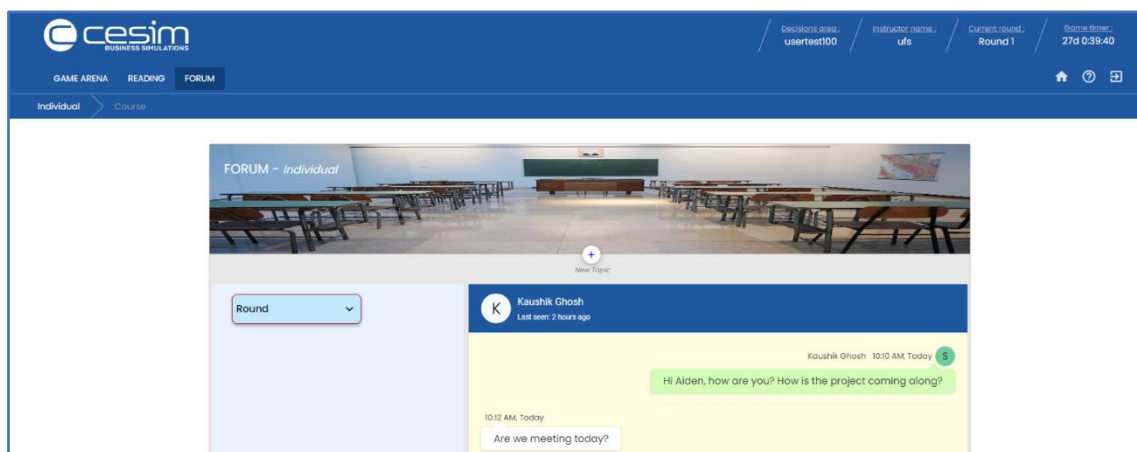


Figure 3 The Forum Tab

- The forums provide an excellent platform for the participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- will receive notifications for messages.

1.4 Main Objective & Winning Criteria

The Ecommerce microsimulation challenges participants to launch and scale a sustainable e-commerce fashion brand, balancing eco-conscious practices with profitability. They will curate a product catalog aligned with sustainability, develop a unique brand identity, and implement digital marketing strategies to attract and retain customers. Additionally, they must optimize the customer journey and manage an efficient, ethical supply chain to stand out in a competitive market.

Success in the simulation depends on achieving key financial metrics, including maintaining an Operating Margin within the optimal range of 20-35%, along with a strong Average Order Value (INR) and an efficient Return on Ad Spend (ROAS). These metrics will reflect participants' ability to drive profitability, optimize marketing investments, and sustain business growth.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

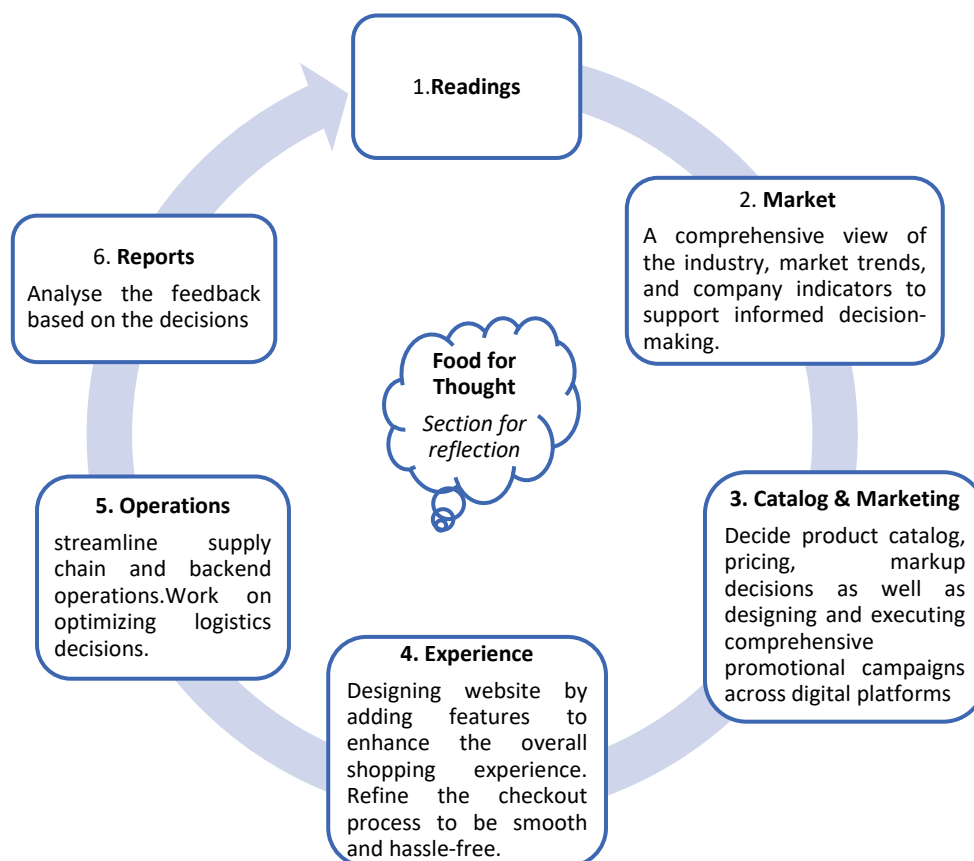


Figure 4 The Sequence of Decision-Making Process

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

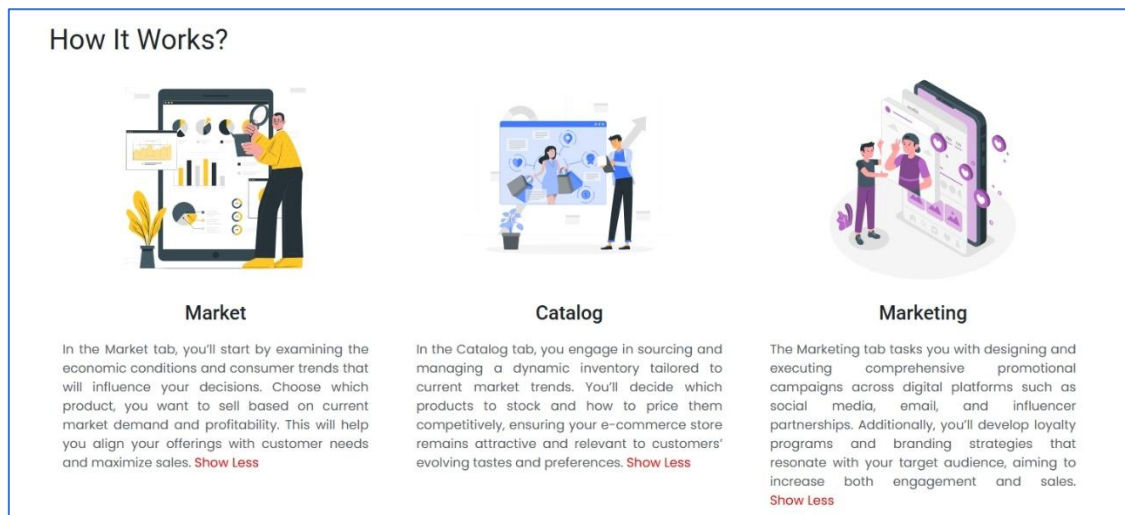


Figure 5 The Introduction Tab in the Game Arena

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.

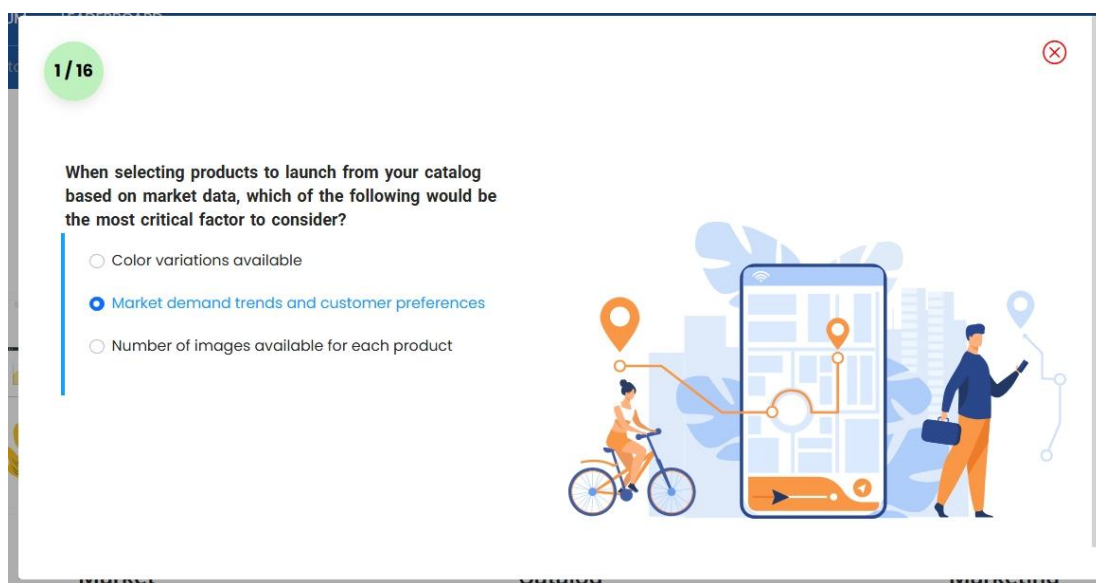


Figure 6 The Food for Thought Section in the Game Arena

2.3 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Memo" section. This section offers vital information on case scenario, and key parameters to consider when making decisions in each tab.



Figure 7 The Market Tab in the Game Arena

2.4 Catalog

Catalog - Product Selection

Participants have to carefully select which sustainable products to feature in their catalog based on market demand, margins, and alignment with their brand positioning. Consider each product's target demographic, raw material sustainability, quality level, and manufacturing impact.

Catalog, Market Information ⓘ							
Selection	Category	Product Name	Target Market ⓘ	Material ⓘ	Quality ⓘ	Sustainability ⓘ	Unit Cost
☒	Women's Apparel	Organic Cotton T-shirt	Urban women, 18-35	Organic cotton	High	Carbon neutral	
☒	Women's Apparel	Bamboo Fiber Blouse	Eco-conscious, 20-40	Bamboo fiber	Medium	Biodegradable	
☒	Women's Apparel	Recycled Polyester Dress	Fashion-forward, 25-35	Recycled polyester	Low	Reduces waste	
☒	Women's Apparel	Hemp Denim Jeans	Casual women, 20-45	Hemp	Very high	Durable	
☐	Women's Apparel	Linen Summer Dress	Summer wear, 18-30	Linen	High	Organic	
☐	Women's Apparel	Modal Fabric Maxi Skirt	Young adults, 18-30	Modal	Medium	Sustainable fiber	
☐	Women's Apparel	Eco-Fleece Hoodie	Winter wear, 18-35	Recycled fibers	High	Reduces landfill waste	
☐	Women's Apparel	Silk Wrap Top	Evening wear, 25-40	Ethically sourced silk	High	Low water usage	
☐	Women's Apparel	Wool Blend Overcoat	Winter wear, 30-50	Wool blend	Very high	Renewable resource	
☐	Women's Apparel	Lyocell Jumpsuit	Fashionistas, 20-35	Lyocell	High	Biodegradable	

Figure 8 The Decision area Under the Catalog Tab

Supplier Selection

Once after deciding the Catalog, the participants are tasked with choosing the right supplier while considering the factors such as quality, sustainability and cost comparison.

Supplier ⓘ

EcoTextile Solutions

Quality (out of 5)	4.5
Sustainability (out of 5)	5
Cost Comparison	Market + 10%

☒

ReGen Fabrics

Quality (out of 5)	3.5
Sustainability (out of 5)	5
Cost Comparison	Market - 10%

☐

GreenFiber Industries

Quality (out of 5)	4
Sustainability (out of 5)	4.5
Cost Comparison	Market

☒

OrganicWeave Ltd.

Quality (out of 5)	4.5
Sustainability (out of 5)	4
Cost Comparison	Market + 4%

☐

PureHemp Co.

Quality (out of 5)	5
Sustainability (out of 5)	5
Cost Comparison	Market + 20%

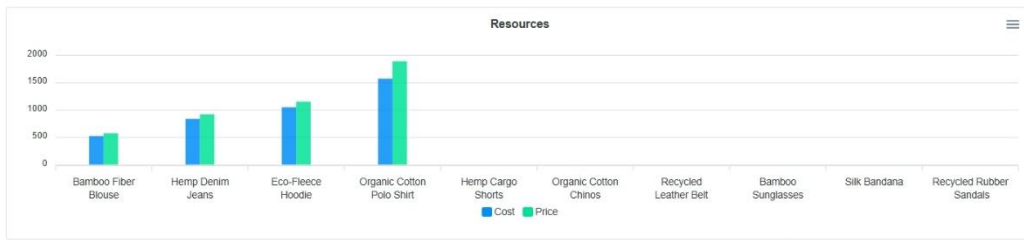
☐

Pricing Decision

Participants are responsible for estimating the monthly sales units for selected products, which will serve as key inputs for projections and budgeting.

Additionally, they must determine appropriate markup percentages for each product by analysing supplier costs, market expectations, and competitor positioning. Finding the optimal markup is crucial—it ensures profitability while maintaining a competitive edge in the eco-conscious market segment.

Pricing ②



Category	Product Name	Supplier Unit Cost, INR	Markups %	Unit Price, INR	Estimate Market Share %	Estimated Monthly Sales, units
Women's Apparel	Organic Cotton T-shirt	525	10	578	3	750
Women's Apparel	Bamboo Fiber Blouse	840	10	924	4	600
Women's Apparel	Recycled Polyester Dress	1,050	10	1,155	5	1,000
Women's Apparel	Hemp Denim Jeans	1,575	20	1,890	6	600

Ordering

Determine proper inventory levels by analysing demand forecasts, seasonality, and stockout risks. Setting appropriate buffer coefficients helps minimize both excess inventory costs and potential lost sales from stockouts.

Ordering ②



Category	Product Name	Monthly Demand, units	Buffer Coefficient	Ordered, units	Standard Shrink %	Safety Stock, units
Women's Apparel	Organic Cotton T-shirt	750	2	1,500	0.5	743
Women's Apparel	Bamboo Fiber Blouse	600	2	1,200	1	588
Women's Apparel	Recycled Polyester Dress	1,000	1	1,000	1.5	0
Women's Apparel	Hemp Denim Jeans	600	1	600	0.8	0

2.5 Marketing

Marketing Strategy: Selecting the Right Campaigns

Under the Marketing tab, the participants are responsible for choosing campaigns that align with the company's overall marketing goals. These campaigns include:

- Facebook Campaigns
- Instagram Campaigns
- YouTube Campaigns
- Email Marketing
- Branding Campaigns

Facebook Campaign ⓘ

 Green Thread Initiative Average reach: 800,000 users Engagement: 5% engagement rate with 15,000 likes and comments Conversion Rate: 2.5% with direct click-throughs to product pages Website Traffic: 20,000 visits The campaign, centered on storytelling, will use posts and short videos to detail the crea... Show More A/B Testing ☐ Cost, INR 60000 ☐	 Eco-Chic Fashion Week Average reach: 1 million users Engagement rate: 3.5% with 25,000 interactions Sales Conversion: 3.2% increase in sales during the campaign week New Followers: 30,000 new followers on social media The campaign is a virtual fashion show that unveils Seamsoco's latest sustainable collecti... Show More A/B Testing ☐ Cost, INR 70000 ☐	 Seamsoco Sustainable Challenge Average reach: 500,000 users Engagement: 10,000 posts under campaign hashtag Follower Growth: 20,000 new followers during the campaign month Engagement Rate: 7% overall interaction rate It invites participants to exclusively wear sustainable fashion for a month, sharing their... Show More A/B Testing ☐ Cost, INR 50000 ☐
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Figure 9 The Decision areas under the Marketing Tab

Designing Targeted Campaigns

Create tailored marketing strategies across Instagram, Facebook, and YouTube to foster community engagement around sustainable fashion. Each platform has unique content needs and varying levels of engagement and conversion potential.

Optimizing Budget Allocation

Distribute the marketing budget strategically across various channels to maximize reach and engagement. Each channel serves different customer segments within the sustainable fashion market, requiring a balanced investment approach.

Branding & Sustainability Narrative

Implement branding initiatives that reinforce your sustainability message. From content marketing and influencer collaborations to experiential events, select campaigns that align with the brand's mission. Each initiative has distinct cost structures and varying impacts on brand awareness and customer acquisition.

By carefully selecting and executing the right marketing strategies, participants can effectively enhance brand visibility and foster a strong, engaged community around sustainable fashion.

2.6 Experience

Under the Experience tab, participants are responsible for designing the website with features that enhance user experience.

To improve engagement and conversion rates, consider adding strategic nudges, such as:

- Scarcity Indicator – Creates urgency by highlighting low stock levels.
- Progress Bar for Checkout – Simplifies the checkout process and reduces drop-offs.
- Dynamic Exit-Intent Pop-Up – Retains visitors by offering incentives before they leave.
- Personalized Recommendations – Enhances shopping experience with tailored product suggestions.
- Social Proof Notifications – Builds trust by showcasing recent purchases or reviews.

Participants can also select the best checkout features to streamline the shopping process and implement an efficient customer service system to enhance support and satisfaction.

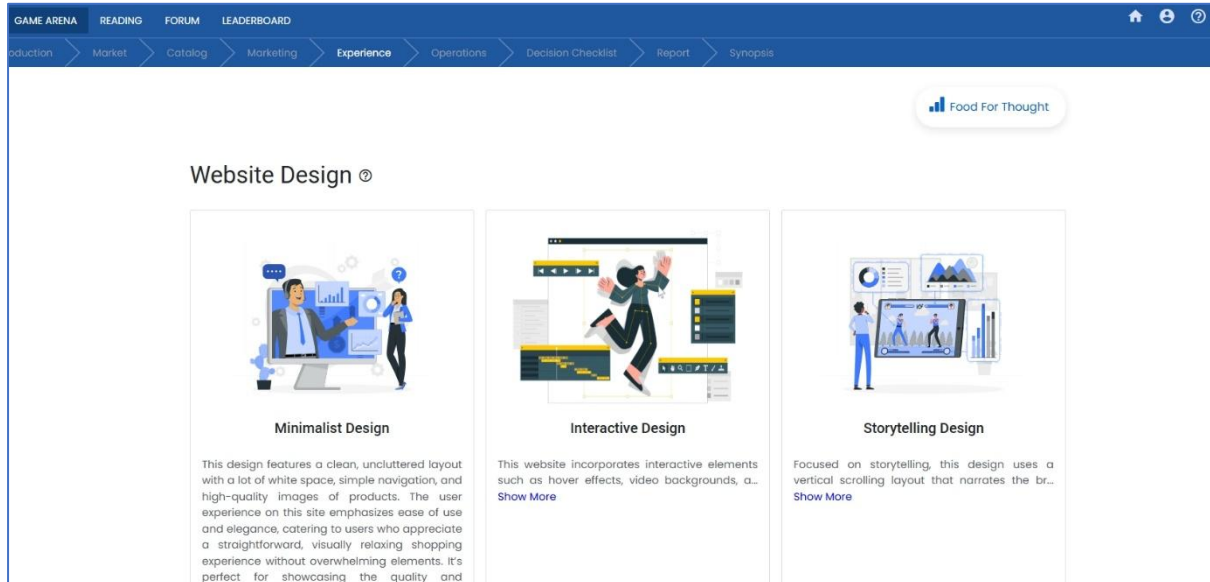


Figure 10 The decision areas under Experience Tab

2.7 Operations

Under the Operations tab, participants are responsible for selecting packaging options that align with both the company's objectives and sustainability goals, while also considering cost implications.

Additionally, choosing the right logistics partner is crucial for streamlining operations. Participants can evaluate various options, each offering distinct advantages based on delivery methods, storage capabilities, reach, and performance metrics.

To further enhance efficiency, participants have access to a range of operational tools, including:

- Vendor Management Software – Optimizes supplier relationships and procurement.
- Advanced Material Quality Testing – Ensures high product standards.
- Blockchain for Supply Chain Transparency – Enhances traceability and trust.
- Dynamic Pricing and Inventory Management Tool – Optimizes pricing strategies and stock control.
- Customer Relationship Management (CRM) Integration – Improves customer interactions and retention.

By making strategic choices in packaging, logistics, and operational tools, participants can create a cost-effective, sustainable, and efficient supply chain.

2.8 Decision checklist

The Decision Checklist assists in reviewing and modifying their decisions before the deadline by consolidating them from each decision areas. Once decisions are finalized, can submit them using the Submit option if they opt to proceed with the Decision checklist.

Decision Checklist	
	Round 2 ▼
	User
Project Portfolio Discount Rate	
Global Outreach Initiative	10%
Regional Domination Campaign	11%
Digital Transformation Drive	12%
Emerging Markets Expansion	0%
Efficiency Overhaul Program	0%
Supply Chain Optimization	0%
Employee Training Enhancement	0%
Automated Claims Processing	0%
Customer Onboarding Redesign	0%
Regulatory Compliance Enhancement	0%
Data Security Strengthening	0%
Innovative Insurance Products	0%
Partnership for New Market Entry	0%

Figure 11 The Decision Checklist

2.9 Report

The report section starts with the ' critical thinking abilities in the food for thought section based on four metrics.

- Rigour: Consistency throughout the decision-making and recommendation.
- Structuring: Logical approach of gathering and analysing information
- Synthesis: Approach towards interpreting information and making informed decisions
- Business Judgement: Ability to evaluate facts, and consider risks and possible implications when making decisions.

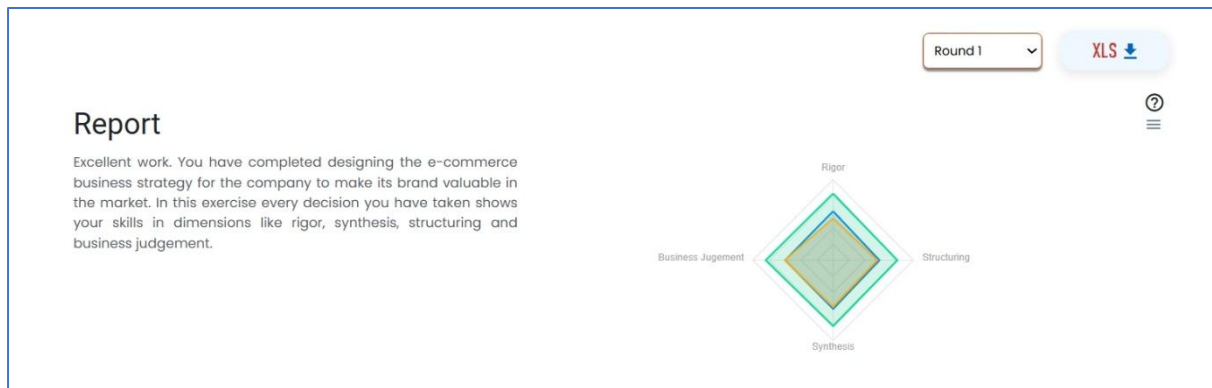


Figure 12 The Report section in the Game Arena

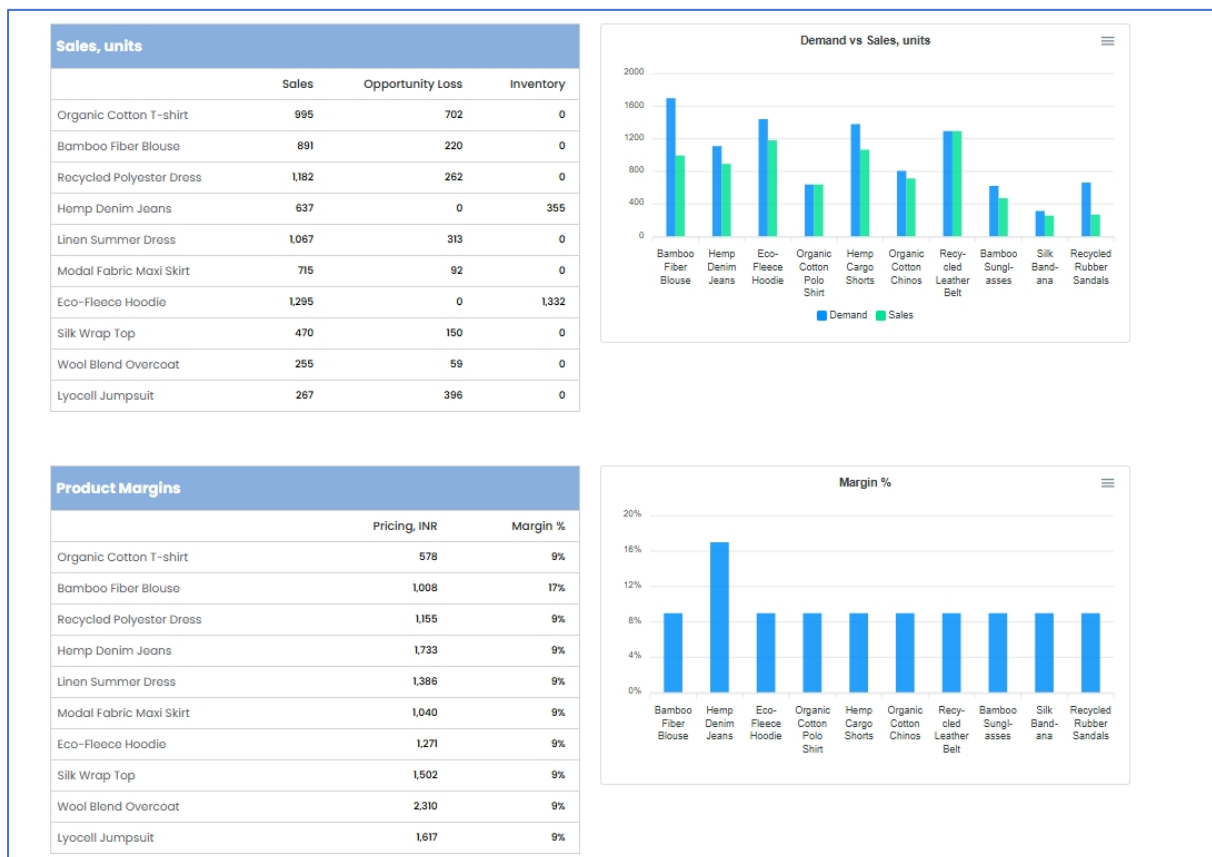


Figure 13 The Report Section in the Game Arena

Key Performance Indicator

Average Order Value (AOV)

- This metric shows the average amount spent per transaction. Higher AOV indicates successful upselling, product mix optimization, and customer willingness to spend more with your brand.
- Sales Performance Indication

This composite score measures overall sales effectiveness across products and channels. A ratio above 1.0 indicates outperforming market averages in converting marketing efforts to actual sales.

- Product Return Rate

The percentage of products returned by customers. A lower return rate suggests good product quality, accurate descriptions, and effective customer expectation management.

- ROAS (Return on Ad Spend)

Measures revenue generated per unit of advertising cost. ROAS above 4.0 indicates highly effective marketing campaigns and efficient use of promotional budget.

Operating Margin %

- Operating margin measures profitability as a percentage of revenue after accounting for variable and fixed costs. A positive and growing operating margin (ideally above 10%) indicates effective pricing strategy and cost control.

Financial Metrics

Revenue

- Total income generated from product sales before deducting costs. Consistent revenue growth (15%+ annually) demonstrates market acceptance and effective customer acquisition strategies.

EBITDA

- Earnings before interest, taxes, depreciation, and amortization reflect operational profitability. Positive EBITDA with growth trend suggests sustainable business operations and effective resource allocation.

Income Statement, k INR	
Revenue	9,538
Cost	8,603
Gross Profit/Loss	935
Promotion cost	694
Website cost	345
Personnel & Service cost	700
Packaging cost	226
Logistics and Inventory cost	659
Process Improvement cost	690
Operating expense	3,314
EBITDA	-2,379

Promotion Distribution		
	Visits	Distribution
Social Media	17,104	34%
Campaigns	19,687	39%
Organic	9,746	20%
Email	3,375	7%

Website Experience	
Bounce Rate	21.7%
Engagement Rate	-2.6%
Customer Satisfaction	73.9%

KPI	
Average Order Value, INR	1227
Sales Performance Indication	0.82
Product Return	5%
ROAS	14
Operating Margin	-25%

Figure 14 The Report Section in the Game Arena

Customer Experience Metrics

- Promotion Distribution

The promotion distribution shows how well the various campaigns through social medias resulted in reaching the target audience and the related percentage.

- Bounce Rate

The percentage of visitors who navigate away after viewing only one page. A lower bounce rate (below 25%) suggests your site is engaging and relevant to visitors' interests.

- Engagement Rate

Measures how actively customers interact with your content and website. A positive engagement rate (above industry average of 2%) indicates compelling content and effective user experience.

- Customer Satisfaction

Percentage of customers reporting positive experiences with your brand. High satisfaction scores (above 80%) correlate with repeat purchases and positive word-of-mouth.

2.10 Synopsis

The synopsis section provides a summary of the decision areas and the report, highlighting how effectively you applied critical thinking in each aspect, outcomes in different areas, offering insights into the impact of your decision-making skills.

Synopsis					
Input		Sales, units			
Parameters	Input	Products	Sales	Opportunity Loss	Inventory
Catalog		Organic Cotton T-shirt	995	702	-
Product 1	Organic Cotton T-shirt	Bamboo Fiber Blouse	891	220	-
Product 2	Bamboo Fiber Blouse	Recycled Polyester Dress	1,182	262	-
Product 3	Recycled Polyester Dress	Hemp Denim Jeans	637	-	355
Product 4	Hemp Denim Jeans	Linen Summer Dress	1,067	313	-
Product 5	Linen Summer Dress	Modal Fabric Maxi Skirt	715	92	-
Product 6	Modal Fabric Maxi Skirt	Eco-Fleece Hoodie	1,295	-	1,332
Product 7	Eco-Fleece Hoodie	Silk Wrap Top	470	150	-
Product 8	Silk Wrap Top	Wool Blend Overcoat	255	59	-
Product 9	Wool Blend Overcoat	Lyocell Jumpsuit	267	396	-
Product 10	Lyocell Jumpsuit				

Figure 15 The Synopsis Section



To play the next iteration, follow these steps:

1. Go to the end of the synopsis page, to find the next round.
2. Click on the next round.
3. The participant will be redirected to the home page.
4. Click on the card to start the next round.

Note: The leaderboard for a round will only be updated only once after moving to the next round.